

healthcare-spending-analysis

March 13, 2026

1 Patterns and Trends in U.S. Healthcare Spending (1996–2016) by Payer, Age, Sex, Care Type, and Disease

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90-800 Exploratory Data Analysis and Visualization with Python

The University of Washington's [Institute for Health Metrics and Evaluation](#) (IHME) evaluates and compiles healthcare data to support and inform healthcare policy and care. Its Disease Expenditure Project analyzes U.S. healthcare spending and provides estimates for 154 conditions across 14 categories, providing insight into spending across public insurance, private insurance, and out-of-pocket payments, as well as health conditions, age group, sex, and type of care.

This dataset incorporates information from government budgets, insurance claims, facility records, household surveys, and other official U.S. sources from 1996-2016. It was released on March 18, 2020, accessed via [Kaggle](#), and downloaded for analysis.

1.1 Data Preparation

The dataset, containing U.S. healthcare spending from 1996 to 2016, was first loaded and cleaned to facilitate analysis. Irrelevant columns were dropped, and key columns were renamed for clarity: `year`, `age`, `care_type`, and `cause`. Combined totals for `age` and `sex` were removed to avoid double-counting, leaving only individual age and sex groups.

Next, categorical variables were recoded to numeric levels for easier plotting and analysis. Age groups were renumbered sequentially, while care types and causes were also assigned numeric codes. Finally, the dataset was checked for missing values, ensuring the dataset was ready for exploratory data analysis.

```
[ ]: # import packages
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

import kagglehub
import shutil

import warnings
warnings.simplefilter(action='ignore', category=FutureWarning)
```

```

# download dataset from kaggle (commented out since dataset already downloaded)
# path = kagglehub.dataset_download("adarshsng/
↳usa-health-care-spending-19962016")

# move dataset into final project working directory
# shutil.move(path, "/Users/miakim/OneDrive/CMU/25 fall/90800/final project")

```

```

[ ]: # read dataset
df_raw = pd.read_csv("2/IHME_DEX_1996_2016.csv")

# select and rename columns
df_clean = (df_raw.drop(columns=["cause_id"])
            .rename(columns={'year_id': 'year',
                              'age_group_id': 'age',
                              'function': 'care_type',
                              'acause': 'cause'}))

# remove combined totals for age and sex
df_clean = df_clean[df_clean["age"] != 22]
df_clean = df_clean[df_clean["sex"] != 3]

# relevel columns
df_clean["age"] = df_clean["age"].replace({28:1,
                                             5:2,
                                             6:3,
                                             7:4,
                                             8:5,
                                             9:6,
                                             10:7,
                                             11:8,
                                             12:9,
                                             13:10,
                                             14:11,
                                             15:12,
                                             16:13,
                                             17:14,
                                             18:15,
                                             19:16,
                                             20:17,
                                             30:18,
                                             160:19})
df_clean["care_type"] = df_clean["care_type"].replace({'AM':1,
                                                         'IP':2,
                                                         'RX':3,
                                                         'ER':4,
                                                         'LT':5,

```

```

'GA':6,
'DV':7})
df_clean["cause"] = df_clean["cause"].replace({'_comm':1,
' _dube':2,
' _inj':3,
' _mental':4,
' _neo':5,
' _neuro':6,
' otherncd':7,
' cirrhosis':8,
' cvd':9,
' digest':10,
' exp':11,
' msk':12,
' resp':13,
' rf':14})

# check for missing values
print('Missing Values:', df_clean.isna().any().any())

# print head
print('df_clean Head:')
print(df_clean.head(5))

```

Missing Values: False

df_clean Head:

	year	age	age_group_name	sex_id	sex	care_type	type_of_care	cause	\
0	1996	2	1 to 4	1	Male	1	Ambulatory care	1	
1	1997	2	1 to 4	1	Male	1	Ambulatory care	1	
2	1998	2	1 to 4	1	Male	1	Ambulatory care	1	
3	1999	2	1 to 4	1	Male	1	Ambulatory care	1	
4	2000	2	1 to 4	1	Male	1	Ambulatory care	1	

	cause_name	mean_all	...	\
0	Communicable, maternal, neonatal, and nutritio...	2575.491802	...	
1	Communicable, maternal, neonatal, and nutritio...	2474.584406	...	
2	Communicable, maternal, neonatal, and nutritio...	2446.218097	...	
3	Communicable, maternal, neonatal, and nutritio...	2395.210613	...	
4	Communicable, maternal, neonatal, and nutritio...	2382.910655	...	

	upper_all	mean_pub	lower_pub	upper_pub	mean_pri	lower_pri	\
0	3124.714039	605.080639	437.147993	778.997570	1650.995720	1296.810196	
1	2863.786701	577.071155	433.332288	714.763493	1582.869154	1358.734096	
2	2761.832455	554.274639	446.150240	657.940286	1580.121390	1399.807654	
3	2643.992850	548.143397	460.727384	649.451470	1544.026671	1362.885139	
4	2626.366456	558.334833	471.963383	648.870566	1528.086866	1341.050495	

upper_pri	mean_oop	lower_oop	upper_oop
-----------	----------	-----------	-----------

0	2060.739834	319.415442	240.206345	410.541398
1	1877.624893	314.644096	258.494447	377.892018
2	1821.051369	311.822065	273.061851	357.326710
3	1737.044092	303.040544	263.238698	341.052978
4	1704.843819	296.488956	253.989528	331.852288

[5 rows x 21 columns]

1.2 Analysis

To prepare the data for analysis by payer type, the dataset was reshaped from wide to long format using a melt operation. The `mean_pub`, `mean_pri`, and `mean_oop` columns, representing public, private, and out-of-pocket spending, were combined into a single spending column with an associated payer identifier. The numeric payer codes were mapped to descriptive labels (“Public,” “Private,” and “Out-of-pocket”) for clarity. Additionally, CI was created to store the lower and upper bounds of the confidence interval as a formatted string; however, these intervals correspond to total spending (`mean_all`) rather than the individual types of payers. This providing an approximate visual representation of uncertainty in spending estimates.

```
[ ]: df_melt = df_clean.melt(
    id_vars=['year', 'sex', 'age', 'age_group_name', 'care_type',
    ↪ 'type_of_care',
    'cause', 'cause_name', 'lower_all', 'upper_all'],
    value_vars=['mean_pub', 'mean_pri', 'mean_oop'],
    var_name='payer',
    value_name='spending'
)

df_melt['payer'] = df_melt['payer'].map({
    'mean_pub': 'Public',
    'mean_pri': 'Private',
    'mean_oop': 'Out-of-pocket'
})

df_melt = df_melt[df_melt['sex'] != 'Both']

df_melt['CI'] = df_melt.apply(lambda row: f"{row['lower_all']} -
    ↪ {row['upper_all']}", axis=1)
```

To summarize spending and associated uncertainty across different groupings, a custom `ci_table` function was created. This function aggregates the reshaped dataset (`df_melt`) by one or more grouping variables, such as year, payer, age, sex, care type, or cause. For each group, it calculates the mean spending and retrieves a single confidence interval (CI) value from the first row of that group. When descriptive labels are preferred, numeric codes are replaced with human-readable names for age groups, care types, and causes. The function is then applied to several combinations of grouping variables, producing tables that provide a concise overview of average spending and approximate confidence intervals across key dimensions of the dataset.

These aggregated tables highlight clear patterns in healthcare spending across key dimensions. Over

time, spending generally increases for all payer types, with private and public spending consistently higher than out-of-pocket costs. Differences between sexes are apparent, with female spending generally lower than male or combined totals. Spending also rises with age, particularly for private and public payers, while out-of-pocket expenditures remain comparatively lower; males tend to have slightly higher spending than females across most age groups. By type of care, ambulatory, inpatient, and dental services account for the largest average expenditures, whereas nursing and emergency care are lower. Finally, examining spending by cause shows the highest expenditures for cardiovascular disease, musculoskeletal disorders, diabetes, and injuries, with lower spending for communicable diseases and liver conditions.

```
[ ]: # confident interval function
def ci_table(group_cols, use_labels=True):
    col_mapping = {
        'age': 'age_group_name',
        'care_type': 'type_of_care',
        'cause': 'cause_name'
    }
    cols_to_use = [col_mapping.get(c, c) if use_labels else c for c in
group_cols]

    table = df_melt.groupby(cols_to_use).agg(
        mean_spending=('spending', 'mean'),
        CI=('CI', 'first')
    ).reset_index()

    print(f"\nConfidence Intervals for {group_cols}:\n", table)

# generate confidence interval tables
ci_table(['year', 'payer'])
ci_table(['year', 'sex'])
ci_table(['age', 'payer'])
ci_table(['age', 'sex'])
ci_table(['care_type'])
ci_table(['cause'])
```

Confidence Intervals for ['year', 'payer']:

	year	payer	mean_spending	CI
0	1996	Out-of-pocket	48.017955	2001.322474 - 3124.714039
1	1996	Private	181.892542	2001.322474 - 3124.714039
2	1996	Public	146.104677	2001.322474 - 3124.714039
3	1997	Out-of-pocket	50.608650	2099.609095 - 2863.786701
4	1997	Private	187.019017	2099.609095 - 2863.786701
..	...	...	...	...
58	2015	Private	381.637208	2100.761762 - 3759.9642
59	2015	Public	343.814258	2100.761762 - 3759.9642
60	2016	Out-of-pocket	78.422865	2044.797666 - 4003.995132
61	2016	Private	399.199023	2044.797666 - 4003.995132

62 2016 Public 353.839056 2044.797666 - 4003.995132

[63 rows x 4 columns]

Confidence Intervals for ['year', 'sex']:

	year	sex	mean_spending	CI
0	1996	Female	147.060358	1602.12206 - 2303.842181
1	1996	Male	103.616425	2001.322474 - 3124.714039
2	1997	Female	152.625648	1650.013923 - 2254.769014
3	1997	Male	106.649042	2099.609095 - 2863.786701
4	1998	Female	160.443339	1705.974819 - 2271.314443
5	1998	Male	111.252847	2205.810705 - 2761.832455
6	1999	Female	168.556523	1651.571035 - 2253.610131
7	1999	Male	116.364148	2166.817782 - 2643.99285
8	2000	Female	178.026096	1633.170036 - 2322.590177
9	2000	Male	122.426767	2140.719276 - 2626.366456
10	2001	Female	190.087279	1706.974503 - 2405.33384
11	2001	Male	130.791904	2159.056555 - 2708.05405
12	2002	Female	204.953350	1817.997973 - 2458.267994
13	2002	Male	141.817711	2198.969153 - 2763.595132
14	2003	Female	218.424003	1874.304943 - 2436.325564
15	2003	Male	151.821507	2200.554831 - 2802.226119
16	2004	Female	227.915457	1916.859725 - 2448.932955
17	2004	Male	158.948089	2182.602885 - 2815.598699
18	2005	Female	235.644983	1995.871676 - 2490.516987
19	2005	Male	164.230582	2198.402019 - 2886.787857
20	2006	Female	244.053881	2126.252324 - 2606.010511
21	2006	Male	169.987127	2285.293484 - 3035.850719
22	2007	Female	250.929012	2257.344682 - 2823.102658
23	2007	Male	175.897680	2442.484672 - 3139.905763
24	2008	Female	255.910496	2457.04495 - 3078.646349
25	2008	Male	179.794654	2599.460164 - 3335.575541
26	2009	Female	264.780859	2627.718 - 3376.348128
27	2009	Male	187.274490	2645.61275 - 3397.440439
28	2010	Female	271.940063	2751.501142 - 3666.025813
29	2010	Male	193.623907	2597.925841 - 3522.874755
30	2011	Female	275.569278	2761.985352 - 3822.871721
31	2011	Male	197.412875	2532.069688 - 3577.683154
32	2012	Female	280.555693	2709.137364 - 3858.910886
33	2012	Male	202.532399	2339.139614 - 3629.760772
34	2013	Female	284.069282	2591.044883 - 3673.390851
35	2013	Male	206.125228	2157.4377 - 3611.474957
36	2014	Female	294.820188	2416.515999 - 3719.852971
37	2014	Male	215.021079	2106.652577 - 3600.9286
38	2015	Female	308.813242	2373.889123 - 4041.50922
39	2015	Male	225.560781	2100.761762 - 3759.9642
40	2016	Female	320.215086	2270.04977 - 4395.475877
41	2016	Male	234.092210	2044.797666 - 4003.995132

Confidence Intervals for ['age', 'payer']:

	age_group_name	payer	mean_spending	CI
0	1 to 4	Out-of-pocket	14.934741	2001.322474 - 3124.714039
1	1 to 4	Private	89.880666	2001.322474 - 3124.714039
2	1 to 4	Public	55.102119	2001.322474 - 3124.714039
3	10 to 14	Out-of-pocket	49.260523	574.7596873 - 736.8563791
4	10 to 14	Private	154.565948	574.7596873 - 736.8563791
5	10 to 14	Public	68.877514	574.7596873 - 736.8563791
6	15 to 19	Out-of-pocket	47.900987	378.320077 - 509.2247855
7	15 to 19	Private	180.614581	378.320077 - 509.2247855
8	15 to 19	Public	84.834014	378.320077 - 509.2247855
9	20 to 24	Out-of-pocket	44.405921	300.144277 - 399.2267391
10	20 to 24	Private	219.574813	300.144277 - 399.2267391
11	20 to 24	Public	108.809501	300.144277 - 399.2267391
12	25 to 29	Out-of-pocket	50.790886	308.7135798 - 406.1799099
13	25 to 29	Private	286.606753	308.7135798 - 406.1799099
14	25 to 29	Public	105.175852	308.7135798 - 406.1799099
15	30 to 34	Out-of-pocket	58.552244	362.847969 - 452.49019
16	30 to 34	Private	349.915069	362.847969 - 452.49019
17	30 to 34	Public	101.522126	362.847969 - 452.49019
18	35 to 39	Out-of-pocket	66.324485	385.3477199 - 501.9323455
19	35 to 39	Private	384.765663	385.3477199 - 501.9323455
20	35 to 39	Public	108.921582	385.3477199 - 501.9323455
21	40 to 44	Out-of-pocket	76.366730	378.4495011 - 496.1389828
22	40 to 44	Private	440.584492	378.4495011 - 496.1389828
23	40 to 44	Public	129.996354	378.4495011 - 496.1389828
24	45 to 49	Out-of-pocket	87.588565	347.4064644 - 480.5918559
25	45 to 49	Private	528.693150	347.4064644 - 480.5918559
26	45 to 49	Public	158.765849	347.4064644 - 480.5918559
27	5 to 9	Out-of-pocket	23.564938	1035.444252 - 1378.588112
28	5 to 9	Private	105.703797	1035.444252 - 1378.588112
29	5 to 9	Public	61.528090	1035.444252 - 1378.588112
30	50 to 54	Out-of-pocket	97.803224	266.5710393 - 401.5502745
31	50 to 54	Private	617.340476	266.5710393 - 401.5502745
32	50 to 54	Public	189.112077	266.5710393 - 401.5502745
33	55 to 59	Out-of-pocket	101.839123	236.3647944 - 335.5341775
34	55 to 59	Private	664.394657	236.3647944 - 335.5341775
35	55 to 59	Public	209.844715	236.3647944 - 335.5341775
36	60 to 64	Out-of-pocket	98.202025	213.9645927 - 309.0674358
37	60 to 64	Private	658.172079	213.9645927 - 309.0674358
38	60 to 64	Public	224.001418	213.9645927 - 309.0674358
39	65 to 69	Out-of-pocket	72.409233	218.2787551 - 326.4954603
40	65 to 69	Private	186.012505	218.2787551 - 326.4954603
41	65 to 69	Public	586.977270	218.2787551 - 326.4954603
42	70 to 74	Out-of-pocket	68.698870	188.8598091 - 292.4418426
43	70 to 74	Private	145.223341	188.8598091 - 292.4418426
44	70 to 74	Public	584.147897	188.8598091 - 292.4418426

45	75 to 79	Out-of-pocket	72.776218	162.2156317 - 257.6430375
46	75 to 79	Private	122.289517	162.2156317 - 257.6430375
47	75 to 79	Public	565.017419	162.2156317 - 257.6430375
48	80 to 84	Out-of-pocket	82.645342	140.6254865 - 239.3972801
49	80 to 84	Private	109.495238	140.6254865 - 239.3972801
50	80 to 84	Public	511.620813	140.6254865 - 239.3972801
51	85 plus	Out-of-pocket	165.707568	129.0948026 - 232.3117368
52	85 plus	Private	150.799376	129.0948026 - 232.3117368
53	85 plus	Public	582.048681	129.0948026 - 232.3117368
54	<1 year	Out-of-pocket	16.743707	935.388907 - 1754.749296
55	<1 year	Private	193.564267	935.388907 - 1754.749296
56	<1 year	Public	96.520958	935.388907 - 1754.749296

Confidence Intervals for ['age', 'sex']:

	age_group_name	sex	mean_spending	CI
0	1 to 4	Female	49.803915	1602.12206 - 2303.842181
1	1 to 4	Male	56.807769	2001.322474 - 3124.714039
2	10 to 14	Female	89.424469	728.1733337 - 967.9080497
3	10 to 14	Male	92.378187	574.7596873 - 736.8563791
4	15 to 19	Female	118.393939	643.5500432 - 905.0875825
5	15 to 19	Male	90.505782	378.320077 - 509.2247855
6	20 to 24	Female	163.338847	636.0922145 - 869.6912576
7	20 to 24	Male	85.187977	300.144277 - 399.2267391
8	25 to 29	Female	206.455925	646.0817641 - 832.4447377
9	25 to 29	Male	88.593069	308.7135798 - 406.1799099
10	30 to 34	Female	234.569820	654.0870898 - 831.8885473
11	30 to 34	Male	105.423140	362.847969 - 452.49019
12	35 to 39	Female	239.815383	672.2435136 - 842.9203468
13	35 to 39	Male	133.525770	385.3477199 - 501.9323455
14	40 to 44	Female	259.532597	646.1961184 - 795.3038897
15	40 to 44	Male	171.765786	378.4495011 - 496.1389828
16	45 to 49	Female	300.371795	591.147026 - 723.0103489
17	45 to 49	Male	216.326581	347.4064644 - 480.5918559
18	5 to 9	Female	59.667696	986.8721383 - 1315.67978
19	5 to 9	Male	67.530187	1035.444252 - 1378.588112
20	50 to 54	Female	338.401374	482.6744746 - 640.8892287
21	50 to 54	Male	264.435811	266.5710393 - 401.5502745
22	55 to 59	Female	353.981962	413.2380661 - 576.5684952
23	55 to 59	Male	296.737036	236.3647944 - 335.5341775
24	60 to 64	Female	346.838434	367.4394663 - 508.4018621
25	60 to 64	Male	306.745247	213.9645927 - 309.0674358
26	65 to 69	Female	300.713299	380.1509064 - 514.8720226
27	65 to 69	Male	262.886039	218.2787551 - 326.4954603
28	70 to 74	Female	289.891953	335.4690105 - 476.8293864
29	70 to 74	Male	242.154786	188.8598091 - 292.4418426
30	75 to 79	Female	290.449567	289.6211745 - 424.3779104
31	75 to 79	Male	216.272536	162.2156317 - 257.6430375
32	80 to 84	Female	290.400378	235.0227051 - 352.1327638

33	80 to 84	Male	178.773884	140.6254865 - 239.3972801
34	85 plus	Female	427.674663	223.307139 - 358.7786074
35	85 plus	Male	171.362420	129.0948026 - 232.3117368
36	<1 year	Female	94.165788	734.0571964 - 1179.593421
37	<1 year	Male	110.386833	935.388907 - 1754.749296

Confidence Intervals for ['care_type']:

	type_of_care	mean_spending \
0	Ambulatory care	385.942629
1	Dental care	456.040829
2	Emergency department care	51.557694
3	Government Administration and Net Cost of Insu...	109.179519
4	Inpatient care	376.227408
5	Nursing care facility	96.904594
6	Prescribed pharmaceutical care	148.917412

CI

0	2001.322474 - 3124.714039
1	126.9635174 - 310.941256
2	266.4520687 - 427.166483
3	294.7010518 - 402.9631542
4	510.0597296 - 718.6723046
5	4.164301104 - 5.73930947
6	370.6820603 - 704.1252246

Confidence Intervals for ['cause']:

	cause_name	mean_spending \
0	Cardiovascular diseases	337.140722
1	Chronic respiratory diseases	150.194479
2	Cirrhosis of the liver	36.579211
3	Communicable, maternal, neonatal, and nutritio...	224.462348
4	Diabetes, urogenital, blood, and endocrine dis...	285.468786
5	Digestive diseases	171.907535
6	Expenditure on risk factors	127.663274
7	Injuries	248.078362
8	Mental and behavioral disorders	196.313884
9	Musculoskeletal disorders	319.088648
10	Neoplasms	140.711201
11	Neurological disorders	185.793342
12	Other non-communicable diseases	228.261397
13	Well care	158.568586

CI

0	60.9733438 - 145.2848351
1	570.9714895 - 888.4880927
2	7.380049622 - 14.60636876
3	2001.322474 - 3124.714039
4	92.09633807 - 295.718846

```

5  109.1019603 - 368.1186401
6   35.93844877 - 155.04885
7  281.6683254 - 593.1205395
8   148.052154 - 297.1924957
9   17.45395551 - 66.17027035
10  56.89077991 - 158.32866
11  62.92362051 - 359.4630289
12  532.1581047 - 770.5071421
13  49.26090069 - 106.9011265

```

Building on the aggregated confidence interval tables, visualizations were employed to further explore patterns in healthcare spending. Line plots were used to track trends over time by payer type, sex, and age group, while boxplots highlighted distributions across care types and causes. These figures complement the summary tables by providing an intuitive, visual perspective on spending dynamics, allowing for easier identification of temporal trends, disparities between subgroups, and variation in expenditures across different types of care and health conditions.

```

[ ]: # Figure 1. Spending Over Time by Year and Payer
plt.figure(figsize=(8,5))
sns.lineplot(data=df_melt, x='year', y='spending', hue='payer', marker='o')
plt.title('Figure 1. Spending Over Time by Year and Payer')
plt.tight_layout()
plt.show()

```

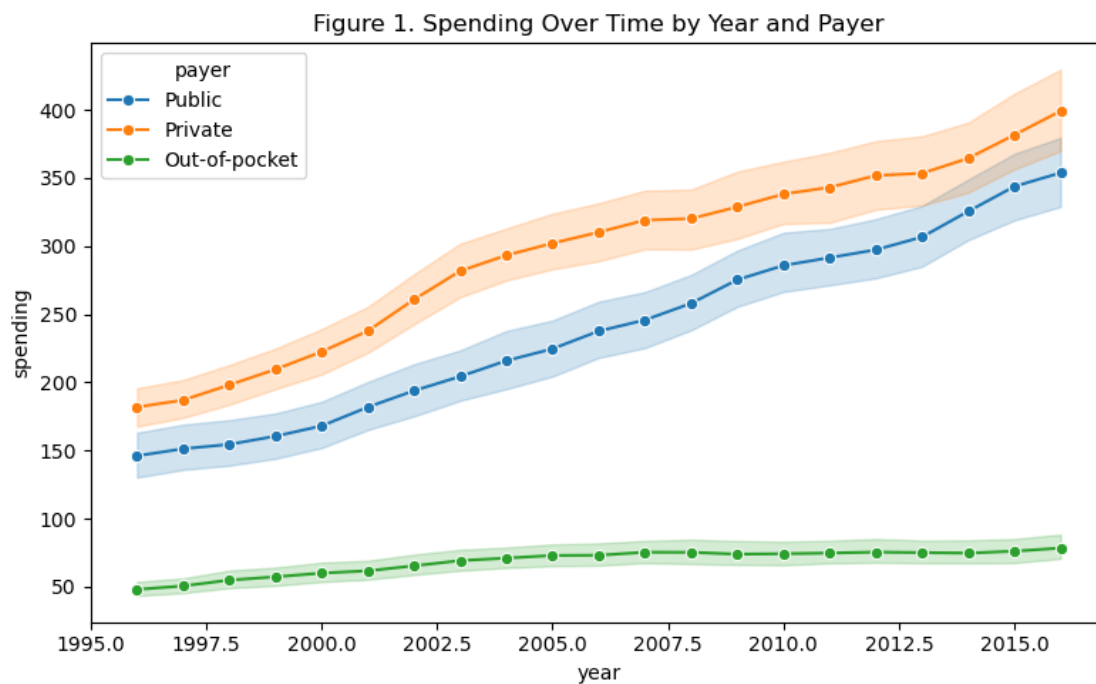


Figure 1. Spending Over Time by Payer This line plot shows mean spending from 1996 to 2016 for public, private, and out-of-pocket payers. Overall, spending has steadily increased for all payer types, with private and public expenditures consistently higher than out-of-pocket. The gap between private and public spending widens slightly over time, highlighting the growing reliance on private insurance to cover healthcare costs. Out-of-pocket spending remains the lowest but shows a modest upward trend, reflecting rising personal financial responsibility for healthcare.

```
[ ]: # Figure 2. Spending by Year and Sex
plt.figure(figsize=(8,5))
sns.lineplot(data=df_melt, x='year', y='spending', hue='sex', marker='o')
plt.title('Figure 2. Spending by Year and Sex')
plt.tight_layout()
plt.show()
```

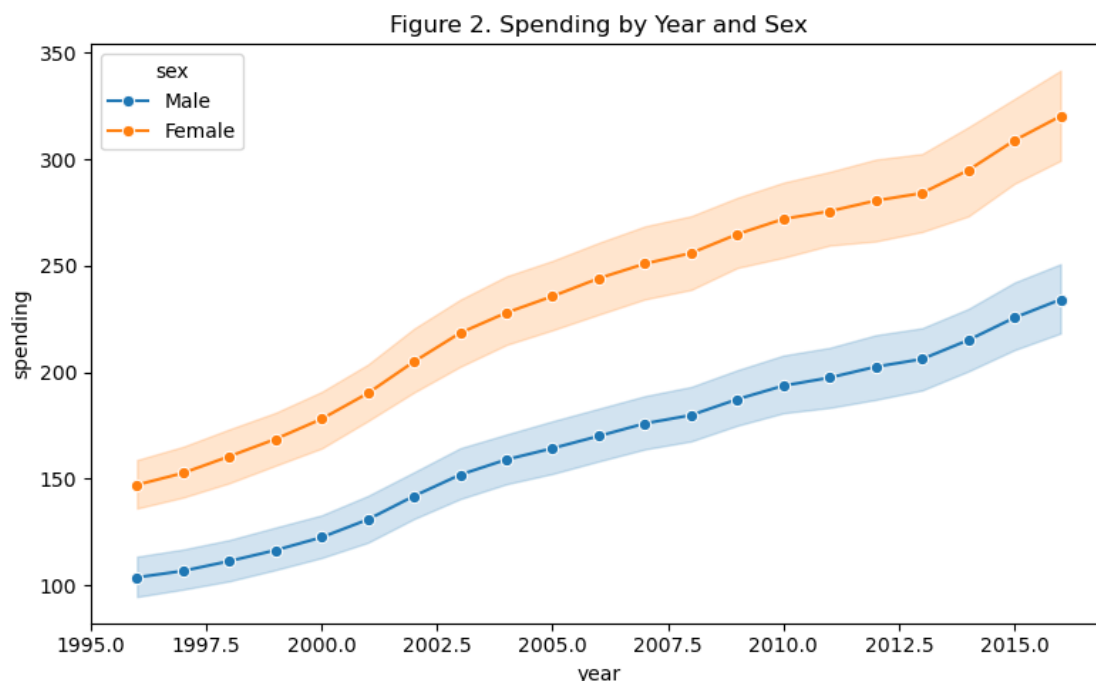


Figure 2. Spending Over Time by Sex This plot illustrates mean spending trends for males and females across years. Males generally have higher spending than females, though both sexes follow similar upward trajectories over the two decades. The gap between sexes is more pronounced in certain years, suggesting demographic differences in healthcare utilization or cost burden. These patterns underscore the importance of considering sex when analyzing healthcare expenditures over time.

```
[ ]: # Figure 3. Spending by Age and Payer
plt.figure(figsize=(10,5))
sns.barplot(data=df_melt, x='age_group_name', y='spending', hue='payer')
plt.title('Figure 3. Spending by Age and Payer')
```

```
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

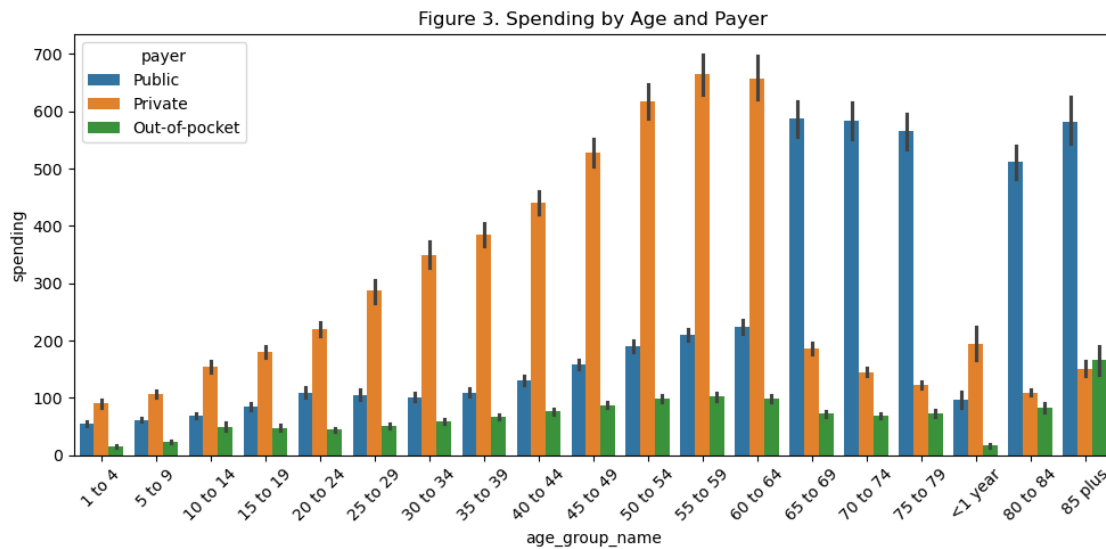


Figure 3. Spending by Age and Payer This line plot highlights how spending varies across age groups for each payer type. Spending increases with age, particularly for private and public payers, reflecting higher healthcare utilization among older adults. Out-of-pocket spending remains relatively low across all age groups. The steepest increases appear in middle to older age brackets, indicating that aging populations drive a substantial portion of healthcare costs.

```
[ ]: # Figure 4. Spending by Age and Sex
plt.figure(figsize=(10,5))
sns.barplot(data=df_melt, x='age_group_name', y='spending', hue='sex')
plt.title('Figure 4. Spending by Age and Sex')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

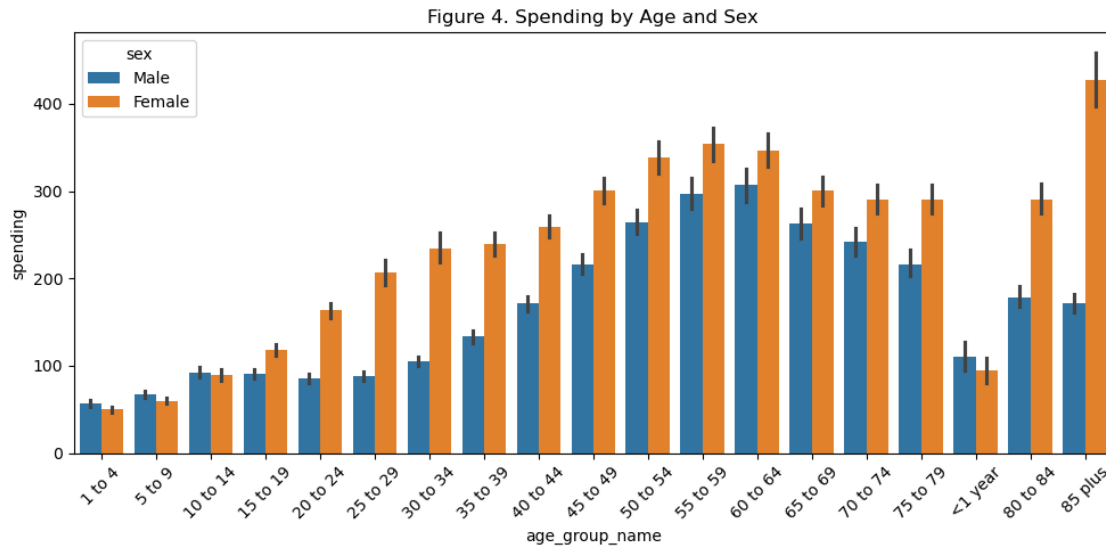


Figure 4. Spending by Age and Sex This plot shows the interaction between age and sex in determining spending. Spending rises with age for both sexes, but males tend to have slightly higher expenditures in most age groups. Differences are particularly noticeable in older age groups, emphasizing that demographic factors, including sex and age, significantly influence healthcare cost patterns.

```
[ ]: # Figure 5. Spending by Care Type
plt.figure(figsize=(10,5))
sns.barplot(data=df_melt, x='type_of_care', y='spending', palette='Set2')
plt.title('Figure 5. Spending by Care Type')
plt.xticks(rotation=90)
plt.show()
```

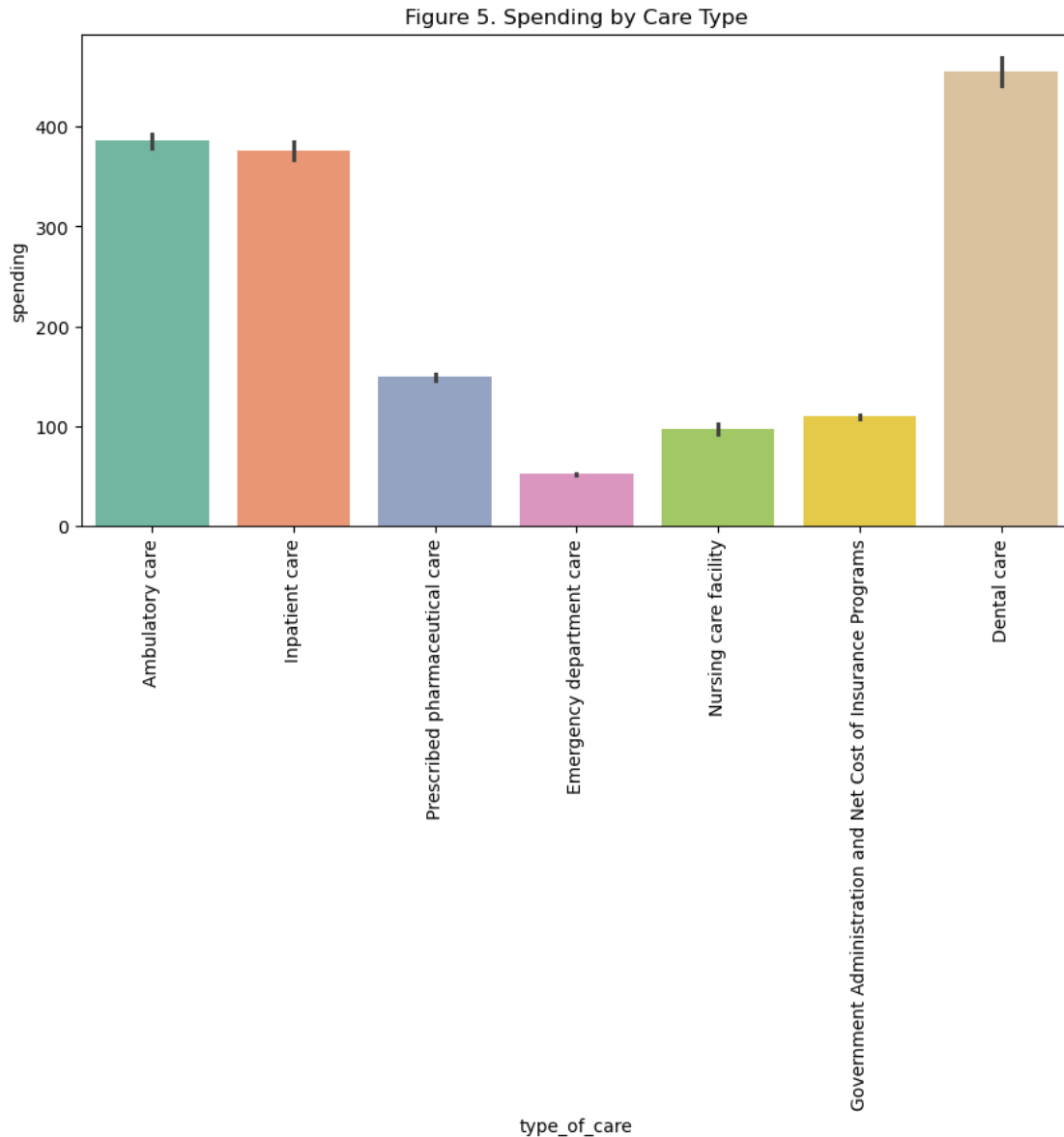


Figure 5. Spending by Care Type The boxplots of spending by care type reveal that ambulatory, inpatient, and dental care account for the largest average expenditures. Nursing, emergency, and pharmaceutical care generally show lower median spending, though some categories exhibit high variability. These distributions highlight the concentration of healthcare costs in specific service types and the variability that exists within each category.

```
[ ]: # Figure 6. Spending by Cause
plt.figure(figsize=(12,5))
sns.barplot(data=df_melt, x='cause_name', y='spending', palette='Set3')
plt.title('Figure 6. Spending by Cause')
plt.xticks(rotation=90)
```

```
plt.show()
```

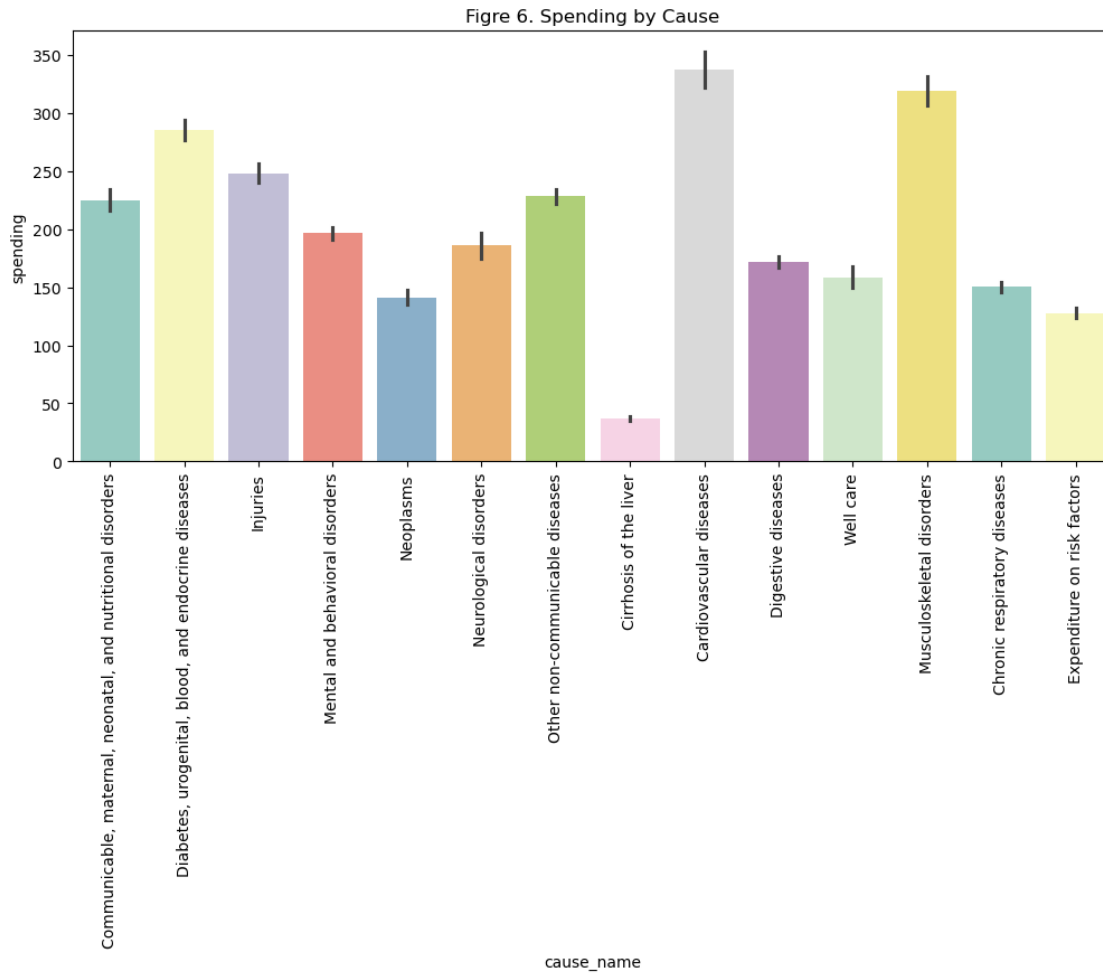


Figure 6. Spending by Cause Boxplots of spending by cause indicate that cardiovascular disease, musculoskeletal disorders, diabetes, and injuries are the leading drivers of healthcare expenditures. Communicable diseases, liver disease, and well care contribute relatively less. The plots also show considerable variability within certain causes, suggesting that individual cases can differ substantially in cost even within the same disease category.

1.3 Conclusions and Discussion

The analysis of U.S. healthcare spending from 1996 to 2016 highlights several clear patterns across payer types, age groups, sexes, care types, and disease categories. Overall, total healthcare spending has increased over time, with private and public expenditures consistently exceeding out-of-pocket payments. Line plots and confidence interval tables both illustrate this upward trend, particularly among older age groups, suggesting that the financial burden of healthcare grows with age and may disproportionately affect public and private payers. Out-of-pocket spending remains comparatively low across most categories, reflecting the structure of insurance coverage and cost-sharing in the

U.S. healthcare system.

Age and sex differences are also evident. Spending generally rises with age, with the highest expenditures observed in older adults for private and public payers. Although male spending is slightly higher than female spending in most age groups, some variation exists across years, possibly reflecting differences in disease prevalence, care utilization, or insurance coverage between sexes.

Analysis by care type shows that ambulatory, inpatient, and dental services account for the largest portions of spending, while emergency and nursing care are lower on average. Cause-specific analyses reveal that cardiovascular, musculoskeletal, diabetes, and injuries dominate spending, consistent with the high prevalence and treatment costs of these conditions. Communicable and liver diseases remain lower in overall expenditure, likely due to lower incidence and more targeted treatment requirements.

Taken together, these findings provide a comprehensive view of healthcare expenditure patterns, highlighting major areas of financial impact and potential policy focus. Subsequent visualizations, such as line plots and bar charts, further illustrate trends over time and across subgroups, allowing for clearer identification of changes in spending patterns and payer responsibilities. These insights can inform policy discussions on resource allocation, preventive care, and the design of interventions aimed at reducing cost burdens for both individuals and the healthcare system. *Disclaimer: This report utilized ChatGPT for text refinement and debugging support.*